Experiment 2: Finite State Automaton in Python

1. Aim

To write a Python program to implement a basic finite state automaton that recognizes strings ending with `'ab'`.

2. Procedure

1. Define the transition table for states `q0`, `q1`, and `q2`.
2. Set the initial state to `q0` and the final accepting state to `q2`.
3. Traverse each character of the input string and update the current state.
4. Check whether the state after processing the string equals `q2`.
5. Display whether each test string is accepted or rejected.

3. Program

```
transitions = {'q0': {'a': 'q1', 'b': 'q0'}, 'q1': {'a': 'q1', 'b': 'q2'}, 'q2': {'a': 'q1', 'b': 'q0'}}

def accepts(s):
    state = 'q0'
    for ch in s:
        state = transitions[state].get(ch, 'q0')
    return state == 'q2'

for w in ['ab', 'aab', 'aba', 'cab']:
    print(w, "->", "Accepted" if accepts(w) else "Rejected")
```

4. Output

```
ab -> Accepted
aab -> Accepted
aba -> Rejected
cab -> Accepted
```

5. Result

Thus, the Python program was successfully executed to implement a basic finite state automaton that matches strings ending with `'ab'`.